

Project3: Final Report

Eduardo Romo

April 17, 2026

Introduction

The Framingham Heart Study is a landmark prospective cohort study initiated in 1948 with 5,209 participants to investigate the causes and risk factors of cardiovascular disease. Participants have been followed biennially with detailed clinical assessments and surveillance for cardiovascular events. The study has been instrumental in identifying key risk factors and characterizing long-term outcomes such as heart disease and stroke. The dataset used for this project is a subset of the Framingham study and includes laboratory, clinical, questionnaire, and adjudicated event data on 4,434 participants collected across three examination periods from 1956 to 1968.

The primary aim of this analysis is to identify baseline risk factors associated with stroke incidence and to estimate the 10-year probability of stroke across different risk profiles, stratified by sex. This will be accomplished using time-to-event methods, specifically Cox proportional hazards models, to evaluate associations between baseline covariates and the hazard of stroke. Statistically, this involves testing the null hypotheses that individual baseline risk factors are not associated with stroke risk (e.g., $H_0: \beta_j = 0$). A secondary aim is to assess how selected risk factors change across the three examination periods. This component will be descriptive and will evaluate patterns in time-varying covariates, including diabetes status and systolic blood pressure, to assess whether more complex modeling

approaches may be warranted. All analyses will be conducted separately for males and females.

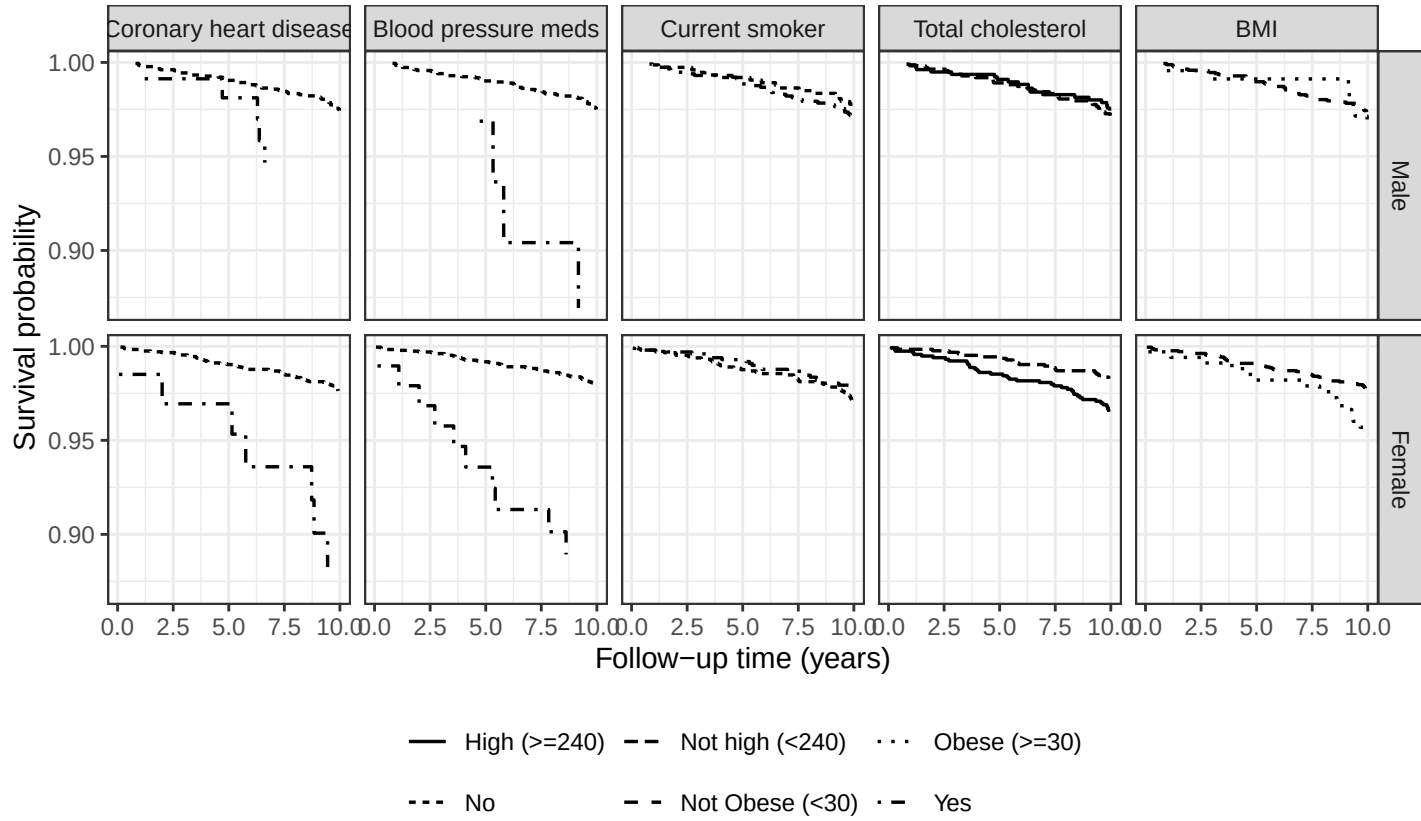


Figure 1: Kaplan-Meier curves for stroke-free survival by sex and baseline risk factors

Table 1: Baseline characteristics overall and stratified by sex.

Variable	Male N=1930	Female N=2472	Overall N=4402
Stroke during follow-up			
No	1881 (97.5%)	2410 (97.5%)	4291 (97.5%)
Yes	49 (2.5%)	62 (2.5%)	111 (2.5%)
Follow-up time (years)	9.40 (1.82)	9.60 (1.54)	9.51 (1.67)
Age	49.73 (8.71)	49.99 (8.63)	49.88 (8.66)
Systolic blood pressure	131.60 (19.40)	133.66 (24.33)	132.76 (22.32)
Blood pressure meds			
No	1868 (97.9%)	2338 (96.1%)	4206 (96.9%)
Yes	40 (2.1%)	96 (3.9%)	136 (3.1%)
Missing, n (%)	22 (1.1%)	38 (1.5%)	60 (1.4%)
Total cholesterol	233.62 (42.39)	239.70 (46.19)	237.01 (44.65)
Missing, n (%)	7 (0.4%)	45 (1.8%)	52 (1.2%)
BMI	26.17 (3.40)	25.57 (4.52)	25.83 (4.08)
Missing, n (%)	4 (0.2%)	13 (0.5%)	17 (0.4%)
Diabetes			
No	1871 (96.9%)	2412 (97.6%)	4283 (97.3%)
Yes	59 (3.1%)	60 (2.4%)	119 (2.7%)
Coronary heart disease			
No	1810 (93.8%)	2405 (97.3%)	4215 (95.8%)
Yes	120 (6.2%)	67 (2.7%)	187 (4.2%)
Current smoker			
No	762 (39.5%)	1471 (59.5%)	2233 (50.7%)
Yes	1168 (60.5%)	1001 (40.5%)	2169 (49.3%)

Table 2: 10-year probability of stroke (Females)

	Age 40	Age 50	Age 60
Baseline	0.006	0.012	0.025
High BP (≥ 160)	0.011	0.023	0.049
Diabetes	0.010	0.021	0.045
High BP + Diabetes	0.020	0.042	0.087
Current Smoker	0.008	0.017	0.036

Table 3: 10-year probability of stroke (Males)

	Age 40	Age 50	Age 60
Baseline	0.006	0.011	0.020
High BP (≥ 160)	0.014	0.027	0.051
Diabetes	0.027	0.050	0.093
High BP + Diabetes	0.067	0.124	0.223
Current Smoker	0.011	0.020	0.038

Table 4: Changes in selected risk factors across examination periods.

Risk Factor	Exam 1	Exam 2	Exam 3
Diabetes, n (%)	121 (2.7%)	155 (3.9%)	254 (7.8%)
Systolic blood pressure, mean (SD)	132.9 (22.4)	136.9 (22.5)	140.2 (22.9)